Recap
U.S. Department of Energy (DOE) Genesis Mission (Fault-Tolerant Quantum Computing) Summit 2026

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Executive Summary

The U.S. Department of Energy (DOE) Genesis Mission Summit 2026 presented a comprehensive vision for the future of American scientific discovery in the era of AI and fault-tolerant quantum computing (FTQC). Government leaders, national laboratory directors, academic researchers, industry executives, and international partners described a coordinated national strategy that extends well beyond the development of individual technologies. Rather than treating AI, high-performance computing (HPC), quantum computing, scientific instrumentation, cloud infrastructure, and advanced networking as separate research domains, the Genesis Mission proposes integrating them into a unified AI-native scientific computing ecosystem designed to fundamentally transform how science is conducted.

A central theme throughout the summit was that the future competitiveness of the United States increasingly depends on computational leadership. Scientific leadership is no longer determined solely by the quality of researchers or the scale of research funding, but by the nation's ability to integrate AI, exascale computing, experimental facilities, scientific data, autonomous software agents, and eventually scientifically relevant fault-tolerant quantum computers into a common computational infrastructure. The Genesis Mission seeks to create precisely such an environment, enabling researchers to accelerate every

stage of the scientific lifecycle—from hypothesis generation and experiment design to simulation, validation, and technology deployment.

The summit also demonstrated that the Genesis Mission is not a conventional Department of Energy program. It has evolved into a whole-of-government initiative, bringing together more than twenty federal agencies, all seventeen DOE National Laboratories, universities, industry partners, and international collaborators around shared scientific challenges. Rather than funding isolated projects, the initiative promotes interoperable computational infrastructure, common AI platforms, shared scientific data ecosystems, and standardized software environments that enable multidisciplinary collaboration at national scale. This organizational transformation is as significant as the underlying technological advances because it redefines how scientific research is coordinated across institutions.

International collaboration emerged as another defining pillar of the Genesis Mission. The announcement of Japan as the first international partner illustrated that future leadership in AI, HPC, and FTQC will depend upon trusted alliances capable of jointly developing computational infrastructure, scientific platforms, and innovation ecosystems. Rather than pursuing isolated bilateral research projects, the United States and Japan committed to building a long-term collaborative framework spanning AI for Science, semiconductors, advanced manufacturing, robotics, fusion energy, and quantum technologies. This partnership reflects a broader recognition that many of the scientific challenges addressed by the Genesis Mission require sustained international cooperation among trusted democratic partners.

The Partnership Panel and platform demonstration translated this strategic vision into operational reality. Industry leaders, national laboratories, and AI startups described how the Genesis platform is already being constructed as a shared computational environment integrating frontier AI models, autonomous AI agents, cloud infrastructure, DOE scientific data, HPC resources, and collaborative software frameworks. Importantly, the platform is designed not as a static software product but as a continuously evolving national capability that improves through contributions from the scientific community. This collaborative architecture establishes a foundation upon which future fault-tolerant quantum computing resources can be incorporated seamlessly into heterogeneous scientific workflows.

The Platform and Portfolio session further expanded this concept by introducing the Genesis Mission as an AI-native scientific operating system. Rather than simply providing researchers with computational resources, the platform captures scientific workflows, AI reasoning processes, software artifacts, validation results, and experimental knowledge to create a continuously learning ecosystem. Concepts such as the AI Flywheel, federated scientific infrastructure, shared AI agents, interoperable workflows, and reusable computational services illustrate how the platform is intended to become progressively more capable as researchers collectively contribute experience and expertise. The resulting positive feedback cycle enables scientific discoveries to improve the platform while platform improvements accelerate future discoveries.

The Congressional Perspectives session broadened the discussion from technical implementation to long-term national strategy. Senators Mike Lee and Mark Warner, together with Under Secretary Darío Gil, emphasized that sustaining American leadership in AI and FTQC will require more than technological innovation. Enduring success depends upon legislative support, educational reform, workforce development, public-private partnerships, trusted governance, resilient energy infrastructure, and sustained national investment. The discussion highlighted the need to transition the Genesis Mission from an Executive Order into long-term statutory authorization while simultaneously preparing the next generation of scientists capable of working alongside increasingly sophisticated AI systems and future logical quantum computers.

From the perspective of fault-tolerant quantum computing, one of the summit's most significant conclusions is that FTQC is no longer being viewed as an isolated hardware objective. Instead, Quantum Genesis positions logical quantum computing as one component within a much larger AI-native scientific computing ecosystem. Future scientific workflows are expected to orchestrate CPUs, GPUs, AI foundation models, autonomous agents, scientific instruments, cloud platforms, exascale supercomputers, and logical quantum processors through intelligent software capable of selecting the optimal computational resource for each task. In this vision, the scientific value of FTQC will arise not merely from achieving larger numbers of logical qubits, but from its seamless integration into heterogeneous computational workflows addressing nationally important scientific challenges.

Taken together, the summit reveals that the Genesis Mission represents a new paradigm for scientific research. It combines mission-oriented AI, exascale HPC, advanced scientific instrumentation, autonomous software agents, cloud infrastructure, federated data ecosystems, and fault-tolerant quantum computing within a unified national platform designed to accelerate discovery across energy, materials science, biology, medicine, climate science, manufacturing, national security, and many other disciplines. Rather than focusing on individual technologies, the Genesis Mission proposes a comprehensive transformation of the nation's scientific enterprise—one in which computational intelligence becomes an active collaborator in discovery and fault-tolerant quantum computing ultimately becomes an integrated component of a continuously learning, AI-native scientific infrastructure.

Part I. Introduction & Technical Keynote: Opening Remarks and Genesis Mission Introduction. Hear from Director Michael Kratsios (OSTP) and Secretary Chris Wright (DOE)

Launching a National Scientific Computing Ecosystem for the AI and Fault-Tolerant Quantum Era
The opening session of the U.S. Department of Energy (DOE) Genesis Mission Summit 2026 established the strategic vision for what may become one of the most ambitious scientific computing initiatives undertaken in the United States since the Apollo Program. Rather than presenting the Genesis Mission as another research funding program or quantum computing initiative, the keynote speakers consistently framed it as a national effort to fundamentally transform how scientific discovery is performed through the convergence of AI, high-performance computing (HPC), advanced scientific instrumentation, and fault-tolerant quantum computing (FTQC). Throughout the opening remarks, speakers emphasized that America has entered a new computing revolution in which computational capability itself has become one of the primary drivers of scientific innovation.

The summit opened by highlighting the remarkable momentum that the Genesis Mission has achieved since its announcement. Organizers described the rapid expansion of partnerships across federal agencies, national laboratories, universities, industry, philanthropic organizations, and international collaborators. Rather than treating these organizations as independent contributors, the Genesis Mission seeks to organize them into a collaborative ecosystem capable of addressing some of the nation's most significant scientific and technological challenges. This collaborative philosophy became one of the defining themes of the keynote: solving next-generation scientific problems requires coordinated national infrastructure rather than isolated research programs.

A recurring message throughout the keynote was that scientific competitiveness increasingly depends upon computational competitiveness. Speakers argued that future breakthroughs in materials science, medicine, clean energy, advanced manufacturing, national security, and fundamental physics will depend not only on scientific expertise but also on the ability to leverage increasingly sophisticated computational resources. The Genesis Mission therefore seeks to provide researchers with integrated access to AI models, exascale supercomputers, quantum processors, scientific data repositories, and the capabilities of all seventeen DOE National Laboratories within a unified computational environment.

A New National Mobilization for Science

Director Michael Kratsios placed the Genesis Mission within the historical context of previous national scientific mobilizations, drawing parallels with landmark initiatives such as the Manhattan Project and the Apollo Program. His remarks suggested that the emergence of AI and advanced computational technologies represents another transformational moment requiring coordinated national investment and strategic vision. Rather than pursuing a single scientific objective, however, the Genesis Mission aims to simultaneously accelerate discovery across multiple scientific disciplines by modernizing the nation's computational research infrastructure.

The keynote emphasized that the Genesis Mission is intended to double the productivity of American scientific research by combining AI-driven scientific reasoning with large-scale computational resources. This vision extends beyond simply deploying larger supercomputers or developing more capable AI models. Instead, it focuses on integrating diverse computational technologies into an intelligent scientific ecosystem capable of supporting researchers throughout the entire discovery lifecycle—from hypothesis generation and experimental design to simulation, validation, and deployment.

AI as a General-Purpose Scientific Accelerator

One of the most compelling aspects of the keynote was the series of scientific lightning talks demonstrating how AI is already transforming multiple disciplines. Rather than presenting speculative future concepts, each speaker described practical examples where AI has begun fundamentally changing scientific workflows.

Under Secretary Darío Gil introduced these presentations by explaining that AI has already demonstrated its ability to accelerate biological discovery through protein structure prediction and related advances. He argued that similar transformations are now emerging across many other scientific fields and that the Genesis Mission exists to accelerate and scale these developments nationally.

The first lightning talk, presented by Fields Medal recipient Professor Terence Tao, explored the rapidly evolving role of AI in mathematics. Professor Tao explained that AI is fundamentally changing mathematical research by accelerating theorem proving, code generation, literature search, proof formalization, and collaborative problem solving. Equally important, he emphasized that AI-generated mathematical results require formal verification to ensure correctness, illustrating how AI and formal reasoning systems can complement one another. His presentation suggested that mathematics is transitioning from an era characterized by proof scarcity to one in which AI may generate proofs faster than humans can analyze them, fundamentally altering mathematical research itself.

The second lightning talk focused on scientific imaging at the DOE Advanced Photon Source. Researchers demonstrated how AI-based reconstruction methods can convert complex X-ray diffraction data into high-resolution three-dimensional images in real time. Tasks that previously required hours or even weeks of computational analysis can now be completed almost instantaneously, enabling scientists to perform experiments interactively rather than waiting for lengthy post-processing workflows. The presentation further illustrated how AI-driven imaging opens the door to autonomous scientific instruments capable of identifying anomalies, adapting experiments dynamically, and accelerating materials discovery.

The third presentation examined one of the nation's most complex engineering systems: the electrical power grid. Researchers described how AI foundation models trained directly on electrical network physics can evaluate billions of operational scenarios, allowing utilities to predict grid behavior far more efficiently than traditional simulation techniques. Rather than replacing physics-based modeling, AI serves as a surrogate model that dramatically accelerates computation while maintaining scientific accuracy. This capability enables planners to evaluate vastly larger design spaces and improve the resilience, reliability, and efficiency of critical infrastructure.

The final lightning talk demonstrated how AI and advanced manufacturing are transforming national security applications. Using an agent-driven design methodology, researchers generated highly optimized flight-test vehicle structures that would have been difficult or impossible for human engineers to conceive independently. Combined with additive manufacturing technologies, these AI-generated designs significantly reduced development time while improving overall system performance. Beyond the specific defense application, the presentation illustrated how AI, simulation, optimization, manufacturing, and scientific expertise can operate together within integrated computational workflows—a recurring theme throughout the summit.

From Individual Technologies to an Integrated Scientific Platform

Following the lightning talks, Darío Gil shifted from illustrating individual scientific successes to describing the broader technical architecture of the Genesis Mission. He explained that the initiative is built upon three interconnected pillars: a portfolio of national scientific challenges, a shared computational platform for accelerating scientific discovery, and educational programs designed to prepare the next generation of computational scientists.

The keynote revealed that the Genesis Mission has expanded from its original twenty-six national science challenges to thirty-three multidisciplinary challenges through collaboration among twenty federal agencies. These challenges now span energy, fusion, electric power systems, materials science, human health, defense, space exploration, and national security. To support these efforts, the DOE announced 278 research awards involving 342 participating institutions distributed across all fifty states, underscoring the national scope of the initiative.

Perhaps the most technically significant portion of the keynote was the description of the Genesis computational platform itself. Rather than viewing AI, classical simulation, HPC, scientific instrumentation, and quantum computing as separate capabilities, the Genesis Mission envisions an integrated scientific infrastructure in which these technologies cooperate seamlessly. The platform combines scientific theory, experimental measurement, national user facilities, exascale supercomputers, AI models, autonomous agents, and emerging fault-tolerant quantum computers into a unified environment capable of orchestrating increasingly complex scientific workflows.

Within this architecture, AI serves multiple complementary roles. It represents scientific knowledge through advanced foundation models, accelerates simulation through surrogate modeling, generates executable code, orchestrates computational workflows using autonomous agents, and assists researchers in connecting diverse scientific resources. Rather than replacing scientists, AI becomes an intelligent collaborator capable of managing increasingly sophisticated computational processes.

Quantum Genesis: Fault-Tolerant Quantum Computing as National Infrastructure

Among the many announcements presented during the keynote, one of the most consequential was the formal introduction of Quantum Genesis, the component of the broader Genesis Mission dedicated to developing scientifically relevant fault-tolerant quantum computing.

Rather than emphasizing raw physical qubit counts or isolated hardware demonstrations, Quantum Genesis focuses on building the first generation of scientifically relevant fault-tolerant quantum computers by 2028 together with user facilities that enable researchers to incorporate logical quantum computation into practical scientific workflows.

This objective closely aligns with the broader DOE vision for scientifically relevant FTQC, which defines success not simply as demonstrating logical qubits but as executing validated end-to-end scientific workloads through tightly integrated AI, HPC, and quantum infrastructures.

The keynote therefore positioned quantum computing not as an isolated technology but as one component within a heterogeneous scientific computing ecosystem. Future scientific workflows are expected to orchestrate CPUs, GPUs, AI systems, HPC clusters, scientific instruments, and quantum processors through intelligent software capable of selecting the most appropriate computational resource for each stage of discovery.

Toward an Internet of Science
Perhaps the most visionary concept introduced during the keynote was the notion of an Internet of Science. Drawing inspiration from the historical evolution of the Internet, Darío Gil proposed extending networked computing beyond digital communication toward interconnected scientific infrastructure.

In this vision, the nodes of the network are no longer merely computers. Instead, they include particle accelerators, synchrotron light sources, telescopes, microscopes, additive manufacturing systems, AI models, HPC facilities, and quantum computers. Autonomous AI agents orchestrate workflows across these distributed resources, enabling researchers to conduct increasingly complex multidisciplinary investigations through a federated computational platform.

This concept represents a profound shift in scientific infrastructure. Rather than transporting data between isolated research facilities, the Internet of Science aims to enable intelligent, automated collaboration among computational resources themselves, dramatically accelerating the pace of discovery.

Building the Next Generation of Scientific Leadership
The keynote concluded by emphasizing that technological infrastructure alone is insufficient. Sustained scientific leadership also requires investment in people. Drawing parallels with the National Defense Education Act following Sputnik, speakers called for a renewed national effort to inspire students to pursue careers in science and engineering. Universities were formally invited to join the expanding Genesis Mission Consortium, reinforcing the initiative's commitment to long-term workforce development and collaborative research.

Overall Perspective
Part I established the intellectual foundation for the entire Genesis Mission Summit. Rather than focusing narrowly on AI or fault-tolerant quantum computing, the keynote presented a far more comprehensive vision: the creation of a national AI-native scientific computing ecosystem in which advanced AI, exascale HPC, scientific instrumentation, and scientifically relevant fault-tolerant quantum computers operate together as an integrated platform for discovery. Within this framework, Quantum Genesis becomes not simply a quantum hardware initiative but a strategic effort to embed logical quantum computing into the nation's future scientific infrastructure, enabling researchers to solve previously intractable problems across chemistry, materials science, energy, biology, national security, and many other domains.

Recording Link: https://www.youtube.com/watch?v=QYbJpLydC7s

Part II. Leadership Panel

A Whole-of-Government Strategy for AI, High-Performance Computing, and Fault-Tolerant Quantum Computing

Following the opening keynote, the second session of the DOE Genesis Mission Summit shifted the discussion from scientific vision to national execution. Whereas Part I established why the Genesis Mission is necessary, the Leadership Panel explored how the United States intends to organize its scientific, technological, and governmental capabilities to achieve this ambitious vision. Leaders representing multiple federal departments and agencies discussed how the Genesis Mission extends beyond the Department of Energy, becoming a coordinated national initiative that leverages expertise, infrastructure, and investments across the federal research ecosystem.

One of the strongest messages emerging from the panel was that the computational challenges confronting modern science no longer align neatly with the organizational boundaries of individual government agencies. Problems such as energy resilience, advanced manufacturing, climate science, biomedical research, space exploration, semiconductor innovation, national security, and next-generation computing all require capabilities that span multiple disciplines. Consequently, the Genesis Mission has adopted a whole-of-government approach, recognizing that meaningful scientific breakthroughs increasingly depend upon coordinated investments in shared computational infrastructure rather than isolated agency-specific programs.

The panel repeatedly emphasized that AI, HPC, and fault-tolerant quantum computing should not be viewed as independent technology initiatives competing for resources. Instead, they represent complementary computational capabilities that collectively form the foundation of future scientific discovery. Each technology contributes unique strengths: AI accelerates scientific reasoning, data interpretation, and autonomous workflow orchestration; HPC continues to provide the large-scale numerical simulation capability required for physics-based modeling; and fault-tolerant quantum computing is expected to accelerate specific computational kernels that remain intractable for classical systems. The Genesis Mission therefore seeks to integrate these technologies into a unified scientific computing ecosystem capable of supporting researchers across every stage of the discovery process.

A central topic of discussion was the importance of shared scientific infrastructure. Rather than requiring every federal agency to independently develop AI models, computational platforms, scientific databases, or quantum capabilities, the Genesis Mission promotes a federated infrastructure in which computational resources can be shared across agencies while remaining aligned with their individual missions. This strategy not only improves efficiency but also enables researchers from different scientific domains to benefit from common computational platforms, standardized workflows, and interoperable software environments.

Panelists also discussed the growing importance of scientific data as a national strategic asset. Modern scientific instruments—including synchrotrons, particle accelerators, fusion experiments, astronomical observatories, biological imaging systems, and advanced manufacturing facilities—generate unprecedented volumes of highly complex data. AI foundation models trained on these scientific datasets have the potential to transform experimental science by accelerating analysis, enabling real-time interpretation, and identifying patterns that would otherwise remain hidden. The Genesis Mission therefore places significant emphasis on integrating scientific data repositories, AI models, HPC resources, and emerging quantum platforms into a cohesive research environment capable of supporting data-intensive scientific discovery.

Another recurring theme was the importance of interoperability. The panel recognized that future scientific workflows will increasingly span multiple organizations, facilities, and computational environments. Researchers may collect experimental measurements at one national laboratory, perform AI-driven analysis on another facility's supercomputer, execute quantum algorithms on specialized fault-tolerant quantum processors, and validate results using additional experimental infrastructure. Supporting such workflows requires common software frameworks, standardized data models, secure computational environments, and intelligent orchestration platforms capable of coordinating heterogeneous resources transparently. This vision closely mirrors the layered AI–HPC–Quantum architecture proposed for scientifically relevant fault-tolerant quantum computing, in which scientific applications, runtime orchestration, quantum control systems, and data infrastructure operate as an integrated platform rather than independent components.

The panel further emphasized that achieving scientific leadership requires more than technological innovation—it also requires sustained public-private collaboration. Industry partners contribute advanced computing platforms, AI technologies, semiconductor manufacturing capabilities, cloud infrastructure, and commercial innovation. National laboratories contribute world-leading scientific expertise and large-scale research facilities. Universities educate future scientists while advancing fundamental research. Government agencies provide long-term strategic direction and mission-oriented investments. The Genesis Mission seeks to unify these complementary strengths within a single collaborative ecosystem capable of addressing national-scale scientific challenges.

An important strategic observation emerging from the discussion was that the Genesis Mission is intentionally application-driven rather than technology-driven. Rather than pursuing AI or quantum computing simply because the technologies are advancing rapidly, the initiative begins with carefully defined scientific grand challenges. Computational technologies are then selected and integrated according to their ability to accelerate progress toward solving these real-world scientific problems. This philosophy represents a significant evolution from earlier technology demonstration programs, placing measurable scientific outcomes at the center of computational innovation.

The panel also highlighted the growing international dimension of scientific leadership. While the Genesis Mission is fundamentally a U.S. initiative, its leaders acknowledged that many scientific challenges—including energy security, climate resilience, advanced materials, public health, and quantum technologies—benefit from collaboration with trusted international partners. Building interoperable scientific infrastructure and fostering collaborative research ecosystems were presented as essential components of maintaining long-term competitiveness while accelerating global scientific progress.

Perhaps the most significant takeaway from the Leadership Panel was its emphasis on institutional transformation. The Genesis Mission is not simply funding individual research projects; it is reshaping how scientific research is organized, coordinated, and executed across the United States. By integrating AI, HPC, scientific instrumentation, cloud infrastructure, advanced networking, and fault-tolerant quantum computing into a unified national platform, the initiative aims to create an entirely new model for computational science—one in which researchers can seamlessly access the most appropriate computational resources regardless of organizational boundaries.

Overall Perspective
Part II demonstrated that the Genesis Mission extends well beyond a single agency or technology program. It represents a national strategy for scientific computing, bringing together federal agencies, national laboratories, universities, industry, and international partners to build a shared AI-native research ecosystem. Rather than viewing AI, HPC, and fault-tolerant quantum computing as separate initiatives, the Leadership Panel presented them as complementary pillars of a unified computational infrastructure designed to accelerate discovery across every major domain of science and engineering. This whole-of-government approach provides the organizational foundation necessary to realize the Genesis Mission's long-term objective: establishing the United States as the global leader in AI-enabled, quantum-enabled scientific discovery.

Recording Link: https://www.youtube.com/watch?v=QVCvh-940Eo

Part III. Fireside Chat : Discussion with His Excellency Shigeo Yamada, Ambassador of Japan

Building an International Alliance for AI, High-Performance Computing, and Fault-Tolerant Quantum Computing

While the first two sessions of the Genesis Mission Summit focused on the national vision and domestic execution strategy, Part III expanded the discussion to the international stage. In a fireside conversation between Dr. Harriet Kung, Acting Deputy Director for Science Programs at the U.S. Department of Energy, and His Excellency Shigeo Yamada, Ambassador of Japan to the United States, the discussion explored why international collaboration has become an essential component of next-generation scientific discovery.

The session marked an important milestone for the Genesis Mission because Japan was officially recognized as the Mission's first international partner. Rather than announcing a traditional bilateral research agreement, both speakers described the partnership as the beginning of a long-term strategic alliance designed to jointly develop AI-driven scientific infrastructure, advanced computing capabilities, and fault-tolerant quantum technologies. Dr. Kung characterized the collaboration as one of the most significant science and technology partnerships ever established between the two countries, supported by a joint commitment of approximately $1 billion over five years.

Unlike many international scientific collaborations that focus on individual research projects, the Genesis Mission partnership was presented as an effort to jointly construct the computational ecosystem that will support scientific discovery over the coming decades. The conversation repeatedly emphasized that AI, high-performance computing, quantum computing, advanced scientific instrumentation, and national research infrastructure are becoming so interconnected that no single nation can maximize their potential through isolated development alone.

AI as the Center of Japan's National Growth Strategy

Ambassador Yamada began by explaining Japan's national AI strategy and why participation in the Genesis Mission aligns naturally with Japan's long-term economic and scientific priorities.

He described AI not merely as another emerging technology but as the centerpiece of Japan's national growth strategy. According to the Ambassador, Japan's government has identified AI and semiconductor technologies as the highest priorities among seventeen strategic sectors that will shape the country's future economic competitiveness and national security. AI is viewed as an essential tool for addressing major societal challenges, including demographic changes, workforce shortages, industrial productivity, healthcare, logistics, and advanced manufacturing.

Particular attention was given to Physical AI, which integrates AI with robotics, autonomous systems, industrial automation, and intelligent machines operating in the physical world. Ambassador Yamada explained that Japan possesses unique advantages in this area through decades of global leadership in robotics, precision manufacturing, and industrial engineering. Equally important, Japanese manufacturing, healthcare, logistics, and service industries have accumulated vast repositories of high-quality operational data that can serve as the foundation for training next-generation AI systems.

The Ambassador also emphasized that Japan has launched its own AI for Science Initiative, recognizing that AI is fundamentally changing every stage of scientific research—from generating hypotheses and designing experiments to analyzing complex datasets and accelerating scientific discovery. Within this context, the Genesis Mission was described as a natural extension of Japan's own scientific priorities, making the partnership between the two countries both strategically and scientifically compelling.

Trust as Strategic Infrastructure

One of the most distinctive aspects of the discussion was the emphasis placed on trust. While scientific capability and technological expertise were acknowledged as essential, Ambassador Yamada argued that the most important reason Japan became the Genesis Mission's first international partner was the extraordinary level of trust that exists between Japan and the United States.

He noted that the two countries are not only longstanding security allies but also deeply integrated economic partners. Japan has been the largest foreign investor in the United States for several consecutive years, and the two nations increasingly collaborate on economic security, advanced technologies, semiconductor manufacturing, AI, quantum computing, and critical infrastructure. This foundation of mutual trust enables deeper collaboration in areas involving sensitive technologies and long-term strategic investments.

The Ambassador argued that, as global competition in emerging technologies intensifies, trusted partnerships become a strategic advantage in themselves. Nations sharing common democratic values, scientific standards, and long-term strategic interests are better positioned to jointly develop transformative technologies while maintaining secure and resilient innovation ecosystems.

Scientific Collaboration Beyond Individual Projects
The discussion highlighted numerous examples of existing collaboration between U.S. national laboratories and Japanese research institutions. Partnerships involving Argonne National Laboratory, Oak Ridge National Laboratory, RIKEN, NVIDIA, Fujitsu, and emerging Japanese fusion companies demonstrate that extensive cooperation already exists across AI, HPC, quantum computing, and fusion energy research.

However, both speakers emphasized that the Genesis Mission represents a significant evolution beyond traditional collaborative research.

Historically, international scientific cooperation has largely been organized around individual projects, specific experiments, or bilateral funding opportunities. The Genesis Mission instead seeks to establish an enduring collaborative infrastructure that continuously supports scientific discovery across many disciplines. Rather than funding isolated investigations, it aims to build persistent computational platforms, shared research environments, common data ecosystems, and interoperable scientific workflows that researchers from both countries can use collectively.

This transition reflects a broader recognition that modern scientific discovery increasingly depends upon shared computational infrastructure rather than isolated experimental facilities.

AI as a Paradigm Shift for International Science
Dr. Kung suggested that AI represents a genuine paradigm shift in scientific research, fundamentally changing how discoveries are generated. Ambassador Yamada strongly agreed with this characterization.

He explained that although Japan and the United States have maintained successful scientific partnerships for decades—including formal science and technology agreements dating back to the 1980s—the emergence of AI requires a fundamentally different model of international cooperation. AI dramatically accelerates scientific research, enabling researchers to move faster, evaluate larger design spaces, automate complex workflows, and integrate enormous scientific datasets. Consequently, scientific collaboration itself must evolve to operate at comparable speed and scale.

According to the Ambassador, participation in the Genesis Mission therefore represents much more than a financial commitment. Both governments have pledged $500 million each over five years, but the larger objective is to create an entirely new collaborative innovation ecosystem connecting governments, national laboratories, universities, industries, startups, and scientific researchers on both sides of the Pacific.

This shift from project-based cooperation toward ecosystem-level collaboration was repeatedly described as one of the defining characteristics of the Genesis Mission.

Toward Global Scientific Leadership

Another important topic discussed during the fireside chat was technological leadership. Ambassador Yamada emphasized that leadership in AI, quantum computing, biotechnology, fusion energy, advanced materials, and related fields cannot be maintained through isolated national efforts alone.

Instead, trusted allies must jointly develop not only technologies but also the standards, research methodologies, computational platforms, and innovation ecosystems that will shape future scientific practice. He argued that collaboration between Japan and the United States should help define the next generation of technical standards while ensuring that emerging technologies reflect shared democratic values and benefit the broader international community.

The conversation therefore extended well beyond bilateral cooperation, framing the Genesis Mission as an opportunity to influence the global trajectory of AI-enabled scientific discovery.

International Collaboration and Quantum Genesis

Although the fireside chat covered AI, semiconductors, biotechnology, robotics, fusion energy, and scientific computing, quantum computing remained one of the strategic pillars of the partnership. Ambassador Yamada specifically identified quantum technologies among the priority areas for future collaboration, reflecting the growing recognition that scientifically relevant fault-tolerant quantum computing will require international cooperation in research, engineering, manufacturing, and software development.

This international perspective aligns closely with the broader objectives of Quantum Genesis, which seeks to build fault-tolerant quantum systems integrated with AI, HPC, and national scientific infrastructure rather than treating quantum computing as a standalone technology. As discussed throughout the summit, practical scientific impact will ultimately depend on heterogeneous computing ecosystems capable of orchestrating multiple computational resources into unified scientific workflows.

Overall Perspective

Part III demonstrated that the Genesis Mission is evolving from a national initiative into an international scientific partnership built upon shared values, trusted alliances, and complementary technological strengths. Rather than pursuing isolated research collaborations, the United States and Japan are jointly investing in the computational infrastructure, scientific platforms, AI capabilities, high-performance computing resources, and fault-tolerant quantum technologies that will define the future of scientific discovery.

Perhaps the most enduring message from the fireside conversation was that scientific leadership in the AI era is no longer achieved solely through technological superiority. It increasingly depends upon trusted international ecosystems capable of combining scientific expertise, computational infrastructure, advanced manufacturing, and long-term strategic investment. Through the Genesis Mission, the United States and Japan are attempting to establish precisely such an ecosystem—one designed not only to accelerate scientific discovery today but also to shape the future architecture of global AI-native, quantum-enabled science for decades to come.

Recording Link: https://www.youtube.com/watch?v=NS25QUkPwBA

Part IV. Partnership Panel & Platform Demo with Genesis Mission Technical Director Brian Spears

Building the National AI Infrastructure for Scientific Discovery
Part IV of the DOE Genesis Mission Summit represented a pivotal transition from strategy to implementation. While the previous sessions focused on national vision, government leadership, and international collaboration, this session addressed a practical question: How will the Genesis Mission actually be built? The discussion brought together leaders from national laboratories, industry, and AI startups to explain how the United States intends to create an integrated scientific computing ecosystem capable of accelerating discovery through AI, high-performance computing, and ultimately fault-tolerant quantum computing. The session concluded with a platform demonstration led by Brian Spears, Technical Director of the Genesis Mission, illustrating that the Mission is already evolving from concept into operational infrastructure.

A recurring theme throughout the panel was that no single organization possesses all of the expertise, infrastructure, or computational resources necessary to transform scientific research at a national scale. Instead, success depends upon combining the complementary strengths of government, national laboratories, universities, startups, and large technology companies into a single collaborative ecosystem. The panelists consistently described the Genesis Mission not as another research program but as a new operating model for scientific innovation, one that deliberately breaks down traditional institutional boundaries.

The discussion opened with examples demonstrating how rapidly this collaborative model can deliver results. Rather than following traditional multi-year procurement cycles, the panel highlighted the rapid deployment of new computational infrastructure through close collaboration among the Department of Energy, AMD, Hewlett Packard Enterprise, national laboratories, and scientific researchers. This emphasis on execution speed reflected one of the defining philosophies of the Genesis Mission: accelerating scientific discovery requires accelerating the development and deployment of the underlying computational infrastructure itself.

Industry as a Strategic Scientific Partner
The panel featured leaders representing emerging AI companies, Google Cloud, and the National Laboratory system. Each speaker emphasized that industry participation extends well beyond providing hardware or cloud services. Instead, commercial partners contribute frontier AI models, scalable cloud infrastructure, specialized computing platforms, engineering expertise, and software ecosystems that can be integrated directly into national scientific research.

Google Cloud, one of the founding industry partners of the Genesis Mission, described the initiative as an opportunity to combine publicly funded scientific knowledge with state-of-the-art AI capabilities. The panel referenced AlphaFold as an example of how publicly available scientific data combined with advanced AI models can fundamentally transform an entire scientific discipline. Rather than viewing AI as merely another research tool, the speakers argued that the Genesis Mission could enable dozens of future breakthroughs of similar significance by systematically connecting scientific data, AI models, and national research infrastructure. Google further announced a $40 million in-kind commitment, including cloud resources, AI models, and large-scale token access through its Gemini platform to support Genesis Mission researchers.

This announcement reflected a broader principle emphasized throughout the panel: the Genesis Mission seeks to democratize access to frontier AI capabilities. Rather than limiting advanced computational resources to a handful of institutions, the Mission intends to provide researchers across participating organizations with access to cutting-edge models, cloud computing resources, and AI-assisted scientific workflows.

National Laboratories as the Foundation of Scientific Infrastructure
Representing the National Laboratory system, SLAC National Accelerator Laboratory Director John Sarrao, serving as Chair of the National Laboratory Directors' Council, described the unprecedented level of collaboration occurring among all seventeen DOE National Laboratories. Rather than operating as independent institutions, the laboratories are working collectively to build a national computational infrastructure capable of supporting the Genesis Mission's ambitious scientific objectives.

A particularly important message from Dr. Sarrao was that the Genesis Mission is expected to improve continuously through iterative collaboration. He described the initiative as "only the tip of the iceberg," suggesting that the current platform represents the beginning rather than the culmination of a much larger scientific transformation. As researchers begin using the platform, their feedback will guide continuous improvements to computational capabilities, software frameworks, scientific workflows, and collaborative infrastructure.

The National Laboratories were also presented as the custodians of some of the world's most valuable scientific assets, including user facilities, experimental instrumentation, scientific datasets, and decades of accumulated research expertise. The Genesis Mission seeks to make these resources more broadly accessible through AI-enabled computational platforms while preserving appropriate protections for sensitive scientific information.

AI as a Force Multiplier for Scientific Discovery
One of the most compelling discussions centered on AI's potential to fundamentally change scientific productivity.

Panelists argued that AI should not be viewed primarily as a labor replacement technology but rather as a scientific force multiplier. Instead of replacing researchers, AI automates repetitive technical tasks, accelerates computational workflows, assists in hypothesis generation, and enables scientists to focus on higher-value intellectual activities.

Misha Laskin, co-founder and CEO of Reflection AI, illustrated this transformation by describing the dramatic improvement of frontier AI models in solving advanced theoretical physics problems. Tasks that once required years of graduate-level research can increasingly be completed with AI assistance in a fraction of the time. His broader point was that AI compresses the time required for scientific discovery, allowing researchers to explore larger scientific design spaces while dramatically increasing research productivity.

Rather than diminishing the importance of scientists, the panel suggested that AI makes scientific careers even more impactful by reducing the time spent on routine computational work and increasing the time available for creative scientific reasoning.

Google Cloud reinforced this perspective through practical examples from national laboratories. One example demonstrated how Gemini AI agents automated electron microscope calibration, reducing manual effort by approximately 90 percent while accelerating setup by a factor of eight. Researchers could therefore devote significantly more time to scientific experimentation rather than equipment preparation.

Measuring Success Through Scientific Impact
An especially important theme throughout the discussion was accountability to the public.

Panelists acknowledged that large national investments in AI infrastructure must ultimately produce tangible benefits for society. Success should therefore be measured not by the size of computational resources deployed but by their impact on real scientific and societal challenges.

Examples discussed included accelerating photovoltaic materials discovery, enabling faster biomedical research, improving electric grid resilience, advancing fusion energy, and shortening the time required to translate scientific discoveries into technologies that improve everyday life. The Genesis Mission was repeatedly described as a mission-oriented initiative, emphasizing practical scientific outcomes rather than technology demonstrations.

This application-driven philosophy closely mirrors the broader objectives of Quantum Genesis, where scientifically relevant fault-tolerant quantum computing is evaluated according to its ability to solve meaningful scientific problems rather than simply achieving larger logical qubit counts.

Collaboration as the Core Architecture

Perhaps the strongest message delivered by the panel was that collaboration itself constitutes one of the Genesis Mission's most important technological innovations.

Rather than treating partnerships among universities, industry, and national laboratories as optional, the Genesis Mission intentionally requires multidisciplinary teams spanning multiple sectors. Panelists explained that many of the funded proposals were assembled within only a few weeks, demonstrating an unprecedented willingness across the scientific community to organize around shared national objectives.

Google described this collaborative framework as an enabling platform that simplifies partnership rather than forcing it. Shared computational infrastructure, common AI platforms, standardized data architectures, interoperable software frameworks, and collaborative governance models make it easier for organizations to contribute expertise while avoiding duplication of effort. The panel further emphasized the importance of open standards, common data models, sovereign data management, and workforce development to ensure that researchers across institutions can effectively collaborate while maintaining appropriate security and governance.

Open and Proprietary AI Working Together

Another significant discussion focused on the relationship between open AI models and proprietary frontier models.

Rather than viewing them as competing approaches, panelists argued that future scientific research will require both. Frontier proprietary models will continue driving advances in reasoning capabilities, while open models will enable customization for specialized scientific domains, reproducibility, and collaborative research. National laboratories similarly emphasized the need for carefully managed research environments capable of balancing scientific openness with appropriate protection of sensitive technologies and national security interests.

This balanced perspective reinforces the Genesis Mission's broader philosophy of heterogeneous computing ecosystems, where multiple computational paradigms coexist and are orchestrated according to the specific requirements of each scientific workload.

Brian Spears: From Vision to Platform

The second half of Part IV shifted from panel discussion to implementation as Brian Spears, Technical Director of the Genesis Mission, introduced the Genesis computational platform.

Addressing the inaugural cohort of challenge teams, Spears described them as the "vanguard" of a new era in American science. He emphasized that the Mission's objective is not simply to provide researchers with AI tools but to fundamentally transform how science and engineering are conducted across the United States.

Spears then introduced what he referred to as the Platform with a capital "P." Rather than a single software application, the platform represents an integrated scientific environment combining frontier AI models, autonomous AI agents, DOE scientific data, experimental facilities, cloud infrastructure, HPC resources, and collaborative software frameworks. AI agents automate specialized scientific tasks, while researchers contribute domain expertise and specialized tools that continuously improve the shared platform.

One of the platform's defining characteristics is its self-reinforcing architecture. Challenge teams are expected not only to use the platform but also to enhance it by contributing new capabilities, identifying missing functionality, and sharing successful workflows. Rather than allowing hundreds of independent teams to develop similar solutions separately, the Genesis technical team integrates these contributions into the shared platform so they become immediately available to the broader scientific community.

Spears also described the significant computational resources already available to participating researchers, including AI models, cloud computing, specialized scientific datasets, AI agents, and infrastructure contributed by AWS, Google, Microsoft, OpenAI, and the Department of Energy. Collectively, these resources exceed $150 million in AI capabilities, providing participating teams with unprecedented computational support from the very beginning of their projects.

The demonstration further introduced foundational platform capabilities, including unified authentication, a Model Access Gateway for accessing frontier AI models across multiple providers, AI agent registries, searchable scientific data catalogs, and standardized access to hundreds of AI agents and DOE-developed models. These capabilities are designed to ensure that researchers across institutions begin with a common set of computational tools while enabling continuous expansion of the platform as the Mission evolves.

Overall Perspective
Part IV marked the transition from vision to execution. Rather than discussing future possibilities, the session demonstrated that the Genesis Mission is already constructing a national AI-native scientific platform integrating government laboratories, universities, industry, cloud providers, frontier AI models, scientific data, autonomous agents, and advanced computational infrastructure into a unified research ecosystem.

For the fault-tolerant quantum computing community, the session offered an equally important insight. The future of scientifically relevant FTQC will not depend solely on advances in quantum hardware. Instead, it will rely on comprehensive computational platforms capable of orchestrating AI, HPC, scientific instrumentation, cloud computing, and, ultimately, fault-tolerant quantum processors within a seamless scientific workflow. The Genesis Mission Platform provides an early blueprint for this integrated future, demonstrating that the next generation of scientific discovery will be driven not by isolated technologies, but by intelligently connected computational ecosystems.

Recording Link: https://www.youtube.com/watch?v=3xdDMopttgE

Part V. Platform + Portfolio

Building the AI-Native Scientific Operating System for the United States

The fifth session of the DOE Genesis Mission Summit marked an important transition from presenting the Genesis Mission platform to engaging directly with the scientific community that will ultimately determine its success. Unlike previous sessions, which emphasized strategic vision, national partnerships, and platform capabilities, this session focused on a fundamental question: How should the scientific community itself shape the future of the Genesis Mission?

Rather than presenting a finished architecture, the Genesis leadership emphasized that the platform remains an evolving national capability that must continuously improve through collaboration with researchers. Scientists participating in the inaugural Genesis Mission projects were therefore invited not merely as users of the platform but as co-designers of its future evolution. This participatory approach represents one of the defining characteristics of the Genesis Mission: the platform is intended to learn alongside its users, becoming progressively more capable as scientific teams contribute new workflows, models, data, tools, and operational experience.

At the center of the discussion was the interaction between two complementary components of the Genesis Mission:

The Platform — the shared national computational infrastructure.

The Portfolio — the nationwide collection of AI-enabled scientific research projects.

Rather than operating independently, the platform and portfolio were presented as a continuously reinforcing ecosystem in which computational infrastructure accelerates scientific discovery while scientific discovery simultaneously improves the computational infrastructure itself.

From Scientific Computing to an Internet of Science

The session began with Rick Stevens describing what he referred to as the Genesis Mission's theory of change.

The objective extends well beyond deploying larger AI models or faster computers. Instead, the Mission seeks to establish American leadership in AI-enabled scientific discovery by creating what Under Secretary Darío Gil previously described as the Internet of Science—an integrated ecosystem connecting AI models, supercomputers, experimental facilities, telescopes, particle accelerators, cloud resources, scientific databases, and researchers into a single computational environment.

Within this vision, AI becomes much more than an intelligent assistant.

Foundation models, domain-specific models, surrogate models, autonomous agents, scientific workflows, laboratory instrumentation, simulation platforms, and scientific data all become interconnected components of a continuously learning scientific system.

Instead of scientists manually coordinating dozens of independent computational tools, the computational infrastructure itself becomes an active participant in scientific discovery.

The AI Flywheel

Perhaps the most technically significant concept introduced during the session was the AI Flywheel.

Rick Stevens argued that the most valuable data in the AI era is no longer simply scientific measurements. Instead, equally valuable is information describing how scientists use AI itself to solve scientific problems.

Every interaction between researchers and AI agents generates rich computational traces:

prompts
reasoning steps
workflow execution
generated software
simulation outputs
experimental planning
validation
scientific artifacts

Rather than discarding these traces, the Genesis platform intends to capture them systematically.

This creates a self-improving feedback loop.

AI agents execute scientific tasks.

Their experiences become training data.

Improved models produce more capable agents.

Those improved agents solve increasingly difficult scientific problems.

Their experiences further improve the platform.

Stevens estimated that the inaugural Genesis portfolio could eventually involve approximately 10,000 AI agents collaborating across hundreds of research projects. Instead of each agent learning independently, all agents contribute to a shared body of computational experience, enabling the entire ecosystem to improve collectively. This continuously reinforcing learning cycle—the "flywheel"—was presented as one of the most important mechanisms through which the Genesis Mission expects to accelerate scientific discovery.

A National Platform Rather Than Thousands of Independent Solutions
Brian Spears expanded on this concept by emphasizing one of the Mission's central engineering principles:

Researchers should stop rebuilding infrastructure that already exists.

Historically, scientific groups have often developed their own software frameworks, AI pipelines, workflow engines, data catalogs, and computational tools independently.

The Genesis platform seeks to eliminate this duplication.

Researchers should instead begin with shared national infrastructure—including frontier AI models, agent frameworks, scientific software, authentication services, and computational resources—and focus their efforts on advancing domain-specific scientific knowledge.

Equally important, Spears stressed that the platform is intentionally incomplete.

Its purpose is not to replace scientific expertise.

Instead, researchers are expected to identify missing capabilities, contribute specialized knowledge, and help improve the platform itself.

This philosophy fundamentally changes the relationship between infrastructure developers and scientific users.

Scientists become co-developers of national computational infrastructure rather than passive consumers of centrally developed software.

Data as the Fuel of Scientific AI

Another major focus of the session was the evolving role of scientific data.

Kelly Rose emphasized that AI-native scientific discovery requires a fundamentally different understanding of data than traditional computational science.

Historically, scientific workflows often followed a relatively linear pattern:

Data collection → Analysis → Publication

The Genesis Mission instead envisions highly iterative workflows where data continuously circulate among experiments, AI models, autonomous agents, simulations, and scientific reasoning.

Scientific data therefore become living computational assets that continuously improve the platform itself.

Rose encouraged participating teams to think about data strategically from the very beginning of their projects.

Every dataset, model output, annotation, validation result, workflow artifact, and computational experiment contributes to a larger scientific data ecosystem capable of accelerating future discoveries.

Rather than viewing data management as an administrative requirement, the Genesis Mission treats it as a foundational component of scientific infrastructure.

Federation: Building One Scientific Instrument from Many Institutions

One of the most forward-looking presentations addressed the concept of Federation.

Rather than centralizing all scientific resources within a single organization, the Genesis Mission seeks to create an interoperable infrastructure that connects many independent institutions while preserving their autonomy.

Federation recognizes the reality of modern science.

The best AI model may exist at one national laboratory.

The most relevant experimental facility may reside elsewhere.

Scientific datasets are distributed across many repositories.

Supercomputers, cloud resources, leadership computing facilities, and edge computing systems operate independently.

Today, scientists themselves must coordinate these distributed resources.

The Genesis Mission aims to transfer this burden from researchers to the computational infrastructure itself.

Importantly, federation was explicitly distinguished from centralization.

The objective is not to create one universal repository or one governing organization.

Instead, federation relies upon:

shared standards
common interfaces
trusted digital identities
interoperable workflows
coordinated governance

allowing independent organizations to collaborate while maintaining institutional independence.

As AI agents increasingly participate in scientific workflows, federation becomes even more important because autonomous systems must operate safely across multiple organizations while respecting security, trust, provenance, and governance requirements.

Building a National Scientific Portfolio
Andy Schwarz then shifted attention toward the Genesis Mission research portfolio itself.

He highlighted the unprecedented national scale of the inaugural program.

The portfolio now includes:

278 research projects
342 participating institutions
95 scientific focus areas
21 national scientific challenge areas

spanning national laboratories, universities, industry, and other research organizations.

Rather than funding isolated investigations, the Genesis Mission intentionally assembled an interdisciplinary national portfolio addressing many of America's highest-priority scientific challenges.

Equally important, project success will be evaluated using two complementary criteria:

scientific excellence

AI innovation

Researchers are expected not only to produce important scientific discoveries but also to demonstrate measurable improvements in scientific workflows through AI-enabled computational methods.

The Genesis platform provides common infrastructure for:

data management
AI model training
inference
workflow sharing
best practices
computational resources

while expecting researchers to contribute feedback that continuously improves these capabilities.

Platform and Portfolio: A Self-Reinforcing Ecosystem
One of the strongest themes throughout the session was that the platform and portfolio are intentionally designed to reinforce one another.

Scientific projects consume platform capabilities.

Their experiences improve the platform.

An improved platform accelerates future scientific projects.

Those improved projects further enhance the platform.

This positive feedback cycle represents one of the most innovative aspects of the Genesis Mission.

Rather than separating infrastructure development from scientific research, both activities become mutually dependent components of a continuously evolving computational ecosystem.

The Community Designs the Platform
Perhaps the most distinctive aspect of Part V was the extensive breakout sessions.

Instead of ending with presentations, the Genesis leadership asked participants to organize into small collaborative groups and answer three fundamental questions:

What capabilities should the platform provide that individual researchers cannot easily build themselves?
What unique capabilities can researchers contribute back to the shared platform?
How should AI agents participate in future scientific workflows?

These questions reveal an important philosophical shift.

The Genesis Mission is not being developed solely by government agencies or software engineers.

Its future architecture will emerge collaboratively through continuous dialogue with the scientific community itself.

Every funded research team becomes simultaneously:

a scientific investigator,
a platform contributor,
an infrastructure evaluator,
and a co-architect of America's future scientific computing ecosystem.
Implications for Fault-Tolerant Quantum Computing

Although this session focused primarily on AI infrastructure rather than quantum hardware, its implications for FTQC are profound.

Future FTQC systems will not operate as isolated quantum processors.

They will become federated computational resources within the broader Genesis scientific platform.

Logical quantum processors will ultimately interact with AI agents, exascale HPC systems, cloud infrastructure, laboratory instrumentation, scientific data repositories, and autonomous experimental workflows through the same interoperable architecture described throughout this session.

The emphasis on shared standards, federation, reusable workflows, continuous learning, AI agents, and self-improving computational ecosystems closely mirrors the software architecture that fault-tolerant quantum computing will require to become a practical component of national scientific infrastructure.

Overall Perspective
Part V demonstrated that the Genesis Mission is not simply building an AI platform—it is constructing an AI-native scientific operating system for the United States.

Rather than viewing AI, data, software, computational infrastructure, and scientific research as separate activities, the Mission integrates them into a continuously learning ecosystem where platforms improve science, science improves platforms, and every discovery contributes to the acceleration of future discoveries.

For the emerging era of fault-tolerant quantum computing, this session offered perhaps the clearest architectural lesson of the summit: future quantum computers will derive their greatest scientific value not as standalone machines, but as seamlessly integrated components of a federated, AI-native national scientific computing infrastructure.

Recording Link: https://www.youtube.com/watch?v=w7CEyEncPAs

Part VI. Congressional Perspectives

Building a National Innovation Strategy for the AI and Quantum Era
The final session of the DOE Genesis Mission Summit elevated the discussion from scientific and technical implementation to national policy. Throughout the day, participants had explored the scientific vision of the Genesis Mission, the computational platform supporting AI-enabled discovery, the integration of high-performance computing and fault-tolerant quantum computing, international partnerships, and the collaborative ecosystem connecting government, academia, national laboratories, and industry. The concluding conversation addressed a broader question: What public policy, legislative framework, and national strategy are required to sustain this transformation over the coming decades?

Moderated by Under Secretary for Science and Innovation Darío Gil, the session featured Senator Mike Lee, Chairman of the Senate committee overseeing the Department of Energy, and Senator Mark Warner, Vice Chairman of the Senate Select Committee on Intelligence. Although the discussion covered AI, scientific research, workforce development, national security, energy infrastructure, and innovation policy, the conversation consistently returned to a common theme: the Genesis Mission represents far more than another federal research initiative. It is emerging as a long-term national strategy designed to strengthen American scientific leadership through AI-native computational infrastructure and future fault-tolerant quantum computing.

Genesis as a National Scientific Enterprise
Darío Gil opened the session by summarizing the extraordinary scale of the Genesis Mission that had unfolded throughout the day.

The inaugural Genesis Mission portfolio includes 278 research teams, selected through the most competitive research solicitation in Department of Energy history from more than 5,000 proposals representing over 800 institutions across the United States. The initiative has expanded from addressing 21 national scientific challenges to 33 grand challenges, supported by participation from 20 federal agencies and departments spanning energy, health, space exploration, national security, advanced manufacturing, and fundamental science. Complementing these public-sector efforts, the Genesis Consortium now includes more than 40 industry partners contributing over $800 million in computational resources, AI technologies, infrastructure, and technical expertise.

This overview reinforced one of the summit's recurring messages: Genesis is not a traditional research funding program. It is an effort to organize the nation's scientific, computational, industrial, and educational capabilities into a coordinated ecosystem capable of accelerating scientific discovery at unprecedented scale.

Innovation Through Incentives Rather Than Central Planning
Senator Mike Lee framed his perspective within the broader constitutional role of government in fostering scientific and technological innovation.

Rather than advocating centralized government direction of scientific research, he argued that the United States has historically succeeded by creating institutions and incentives that encourage private innovation, entrepreneurship, and discovery. Patent systems, intellectual property protections, competitive markets, and targeted public investments have enabled researchers and inventors to transform scientific ideas into practical technologies while maintaining an environment that rewards creativity and risk-taking.

Within this context, Senator Lee described the Genesis Mission as a mechanism for improving the effectiveness of existing national investments rather than expanding government control over scientific research. The United States already invests roughly one trillion dollars annually in research and development across government, industry, universities, and philanthropy. The central objective of Genesis, he suggested, is to dramatically increase the productivity of those investments by providing AI-native computational infrastructure that enables researchers to accomplish more with existing resources.

This perspective aligns closely with one of the summit's overarching themes: AI and future fault-tolerant quantum computing should be viewed as productivity multipliers for science, enhancing the nation's existing research enterprise rather than replacing it.

Education as National Infrastructure
A substantial portion of the discussion focused on scientific workforce development.

Senator Mark Warner argued that maintaining leadership in AI and quantum technologies requires not only advanced computing infrastructure but also a dramatic expansion of the nation's scientific talent pipeline. Drawing inspiration from the National Defense Education Act of 1958, enacted following the launch of Sputnik, he proposed accelerating the education of American scientists and engineers by creating new pathways allowing undergraduate and doctoral education to be completed more efficiently.

Darío Gil expanded on this historical comparison by noting that the National Defense Education Act increased American production of science and engineering PhDs by approximately ten percent annually over an extended period. He suggested that the AI era may require a comparable national educational initiative—what he described conceptually as a National Defense Education Act 2.0—designed to prepare a new generation of scientists capable of working effectively alongside AI systems, high-performance computing, and future quantum technologies.

The discussion highlighted a broader shift in thinking. Scientific leadership is no longer determined solely by research funding or laboratory infrastructure. Increasingly, it depends upon cultivating a workforce capable of leveraging advanced computational tools to accelerate discovery across every scientific discipline.

From Executive Order to Enduring National Policy
Another major topic addressed the long-term governance of the Genesis Mission.

The initiative was originally launched through an Executive Order, enabling rapid implementation and coordination across federal agencies. Both Senators emphasized, however, that scientific programs intended to span decades require a more durable legislative foundation. Executive actions can change with administrations, whereas statutory authorization provides continuity, stability, and long-term confidence for researchers, universities, industry, and international partners.

Senator Lee therefore argued that the Genesis Mission should ultimately be codified through legislation, transforming it from an executive initiative into a permanent national capability. Senator Warner agreed while emphasizing that such legislation must preserve scientific integrity through peer review, sustained investment in basic research, and bipartisan support insulated from short-term political fluctuations.

This discussion underscored that transformational scientific infrastructure must be supported not only by technological innovation but also by institutional continuity.

AI as an Accelerator of Scientific Productivity
Returning to the central scientific objectives of the Genesis Mission, Darío Gil argued that AI is fundamentally changing the economics of scientific research.

Examples presented throughout the summit demonstrated AI dramatically accelerating mathematical reasoning, materials science, microscopy, engineering design, and scientific simulation. Rather than simply automating existing workflows, AI enables scientists to explore larger design spaces, test more hypotheses, and solve problems that previously required decades of effort. Gil suggested that the Genesis Mission aims to double the productivity of American research and development without requiring proportional increases in spending, fundamentally altering the rate at which scientific discoveries can be translated into societal benefits.

This framing reinforces the Genesis philosophy that computational capability should ultimately be measured by scientific impact rather than raw computational performance.

Balancing Innovation and Responsible Governance
One of the session's most nuanced discussions concerned the societal implications of rapidly advancing AI.

Senator Warner expressed strong support for AI-driven scientific innovation while cautioning that the pace of technological progress may outstrip society's ability to adapt. He raised concerns regarding workforce disruption, cybersecurity, autonomous AI agents, frontier model governance, and the need for safety evaluations prior to deployment of increasingly capable AI systems.

In contrast, Senator Lee emphasized that although AI will inevitably transform many occupations, technological revolutions throughout history have generally created new forms of employment while expanding human productivity. He argued that AI should be viewed primarily as a tool that augments human capabilities rather than replaces them, citing examples from biomedical research and legal practice where AI enhances professional effectiveness without eliminating the need for expert human judgment.

Although approaching the issue from different perspectives, both Senators agreed that AI's benefits will depend upon thoughtful governance, public trust, and sustained investment in education and workforce development.

Purpose-Driven AI
Perhaps the most compelling message of the session came during Darío Gil's closing remarks.

He observed that much of the public discussion surrounding AI has become abstract, focusing on speculative future scenarios rather than tangible societal benefits. The Genesis Mission intentionally adopts a different philosophy.

Rather than developing AI for its own sake, Genesis organizes AI around purpose.

Its objective is to accelerate fusion energy, design advanced materials, discover new medicines, strengthen national security, modernize manufacturing, improve climate resilience, and solve scientific problems that directly affect society. By grounding AI development within clearly defined scientific missions, the initiative seeks to demonstrate why these technologies matter to citizens while ensuring that technological progress remains aligned with national priorities.

This emphasis on mission-oriented AI distinguishes the Genesis Mission from many purely commercial AI initiatives, reinforcing the role of public scientific institutions in directing advanced computational capabilities toward long-term societal challenges.

Energy as the Foundation of the AI Era
The discussion concluded by highlighting one resource underpinning every aspect of future computational infrastructure: energy.

Both Senators emphasized that AI, exascale computing, and future fault-tolerant quantum computing will require unprecedented amounts of reliable electrical power. They identified advanced nuclear technologies, small modular reactors, fusion energy, geothermal systems, and other firm power sources as essential components of America's future innovation ecosystem. Reliable, scalable, low-carbon electricity was presented not simply as an energy policy issue but as foundational infrastructure for scientific leadership in the AI and quantum era.

This perspective reinforces an important systems-level insight: scientific computing, AI, HPC, and FTQC ultimately depend upon advances across multiple domains, including energy generation, semiconductor manufacturing, communications infrastructure, and workforce development.

Implications for Fault-Tolerant Quantum Computing
Although the final panel focused primarily on AI and national innovation policy, its conclusions have profound implications for fault-tolerant quantum computing.

The discussion repeatedly emphasized that FTQC should not evolve as an isolated technology. Instead, it must become part of a broader national scientific infrastructure integrating AI, exascale HPC, advanced networking, scientific instrumentation, and mission-driven research. Long-term success will therefore depend not only on achieving logical qubits and error correction milestones, but also on sustained public investment, enduring legislative support, robust educational programs, trusted public-private partnerships, and resilient energy infrastructure.

In this context, the Genesis Mission provides a policy framework that complements the technical roadmap for FTQC by recognizing that transformative quantum capabilities require equally transformative institutional, educational, and societal foundations.

Overall Perspective
The closing session demonstrated that the Genesis Mission is as much a national innovation strategy as it is a scientific computing initiative. Throughout the summit, AI, high-performance computing, and future fault-tolerant quantum computing were consistently presented as enabling technologies whose ultimate value lies in accelerating scientific discovery and improving society.

The final conversation broadened this vision by emphasizing that sustained leadership in the AI and quantum era will require coordinated action across research institutions, industry, government, education, legislation, and infrastructure. If earlier sessions defined how AI-native scientific discovery will be achieved, the Congressional Perspectives panel addressed how the nation itself must evolve to support that transformation over the coming decades.

Recording Link: https://www.youtube.com/watch?v=RUEcupiHmtE

AI-NATIVE QUANTUM COMPUTING ACADEMY

https://www.linkedin.com/company/115996111